SE-NAV — Software Engineering (CS-326) — Master Index

Weeks 1–15 · 17 Aug – 29 Nov 2026 · **Midterm 25** / **Final 35** / **Lab 40** · load 2.5 · difficulty 2.5/5

> **Open this file first.** It is the map: where every topic lives, every line offset, every resource. **Four-file pack** (all in `terms/2026-fall/SE/`): | File | Holds | |---|---| | `SE-NAV.md` | This map — glance tables, master offsets, resources index | | `SE-BEFORE-MID.md` | Weeks 1–8 (midterm) — full per-week content | | `SE-AFTER-MID.md` | Weeks 9–15 (final) — full per-week content | | `SE-LABS.md` | All 13 labs + lab viva — lab→week map, per-lab blocks, viva prep |

Where to look (fast answers)

I want…	Open
A topic with its defs/formulas/numericals	that week's block in BEFORE-MID / AFTER-MID
Any week's exact line offsets	**Master offset table** below
A full fear-killer pack (verbatim)	that week's block — packs carry per-question offsets
A lab, step-by-step	`SE-LABS.md` → per-lab block
Viva questions	`SE-LABS.md` viva section · `Viva-Book.md`
Books / YT playlists / MOOCs / practice banks	**Resources** section in this file
Exam-day order	**Exam stacks** section
Which topic covers what	**Topic map** section

Course facts

- **Code:** CS-326 · **Credits:** 3+1 (theory + lab) · **Contact:** 3 lectures + 1 lab/week · **Prereqs:** CS-222 (OOP), CS-201 (Data Structures)
- **Weightage:** Midterm 25% / Final 35% / Lab 40% (official, per `Week-by-Week-Narrative.md` L11). Study-effort split is ~80% theory / ~20% lab (`Marks-Allocation.md`) — **both are kept, labeled**; the 40% lab is the official risk component.
- **Difficulty:** 2.5/5 · **GPA Risk:** Medium (group project) — *"the group project is where GPAs go to die"* (narrative L3)
- **Rotation:** Course B (Thu + Fri) — 2 B-slots/wk (A = CCN Thu, AI Fri); no Saturday lab
- **Midterm:** Week 8 (05–11 Oct) · **Final:** Week 15 (23–29 Nov) · **Lab viva:** Week 15 (separate slot)
- **Exam character:** memorization-heavy — frameworks, models, terminology. Theory can get a B; the group project decides A vs C.
- **Examinability note:** no `02-Official-Syllabus.md` — weightage/examinability sourced from `01-Course-Overview.md` + `Marks-Allocation.md` only (flagged, not fabricated).

Master weekwise line-offset table

Offsets are line numbers into the named source file. `Fear-Killer-Packs.md` and `Week-by-Week-Narrative.md` are the two anchored sources; `weeks/SE-Wn.md` files live in `weeks/`. SE packs are **topic-keyed and lag narrative weeks** (pack Week 3 = project mgmt lands in narrative W4, etc. — see `weeks/README.md`). W3, W7, W8, W14 have no pack.

Week	Narrative	Pack heading	Pack questions	weeks/SE-Wn	Topic
W1	L17–35	L11 (W1: process-models)	res L12 · Q1 L14 · Q2 L15 · Q3 L16	SE-W1 L1–51	Process models
W2	L38–54	L18 (W2: requirements-engineering)	res L19 · Q1 L21 · Q2 L22 · Q3 L23	SE-W2 L1–50	Requirements engineering
W3	L57–74	— (no pack: DFD/ER)	—	SE-W3 L1–50	Analysis modelling
W4	L77–93	L25 (W3: project-management)	res L26 · Q1 L28 · Q2 L29 · Q3 L30	SE-W4 L1–51	Project management

W5	L96–112	L32 (W4: software-design)	res L33 · Q1 L35 · Q2 L36 · Q3 L37 · Q4 L38	SE-W5 L1–49	Software design & architecture
W6	L115–133	L40 (W5: agile-and-scrum) + L32 Q3	res L41 · Q1 L43 · Q2 L44 · Q3 L45	SE-W6 L1–51	Design patterns & agile
W7	L136–152	— (revision, no pack)	—	SE-W7 L1–44	Midterm revision
W8	L155–162	— (no pack)	—	SE-W8 L1–31	MIDTERM
W9	L165–184	L47 (W6: verification-and-validation)	res L48 · Q3 L52	SE-W9 L1–52	V&V (recovery)
W10	L187–204	L47 (W6: V&V)	Q1 L50 (cyclomatic)	SE-W10 L1–45	White-box testing
W11	L207–224	L47 Q2 + L62 (W8: maintenance-and-sqa) Q3	Q2 L51 · Q3 L67	SE-W11 L1–48	Black-box testing & SQA
W12	L227–251	L54 (W7: cost-estimation)	res L55 · Q1 L57 · Q2 L58 · Q3 L59 · Q4 L60	SE-W12 L1–54	COCOMO
W13	L254–269	L62 (W8: maintenance)	res L63 · Q1 L65 · Q2 L66	SE-W13 L1–47	Software maintenance
W14	L272–293	— (taper, no pack)	—	SE-W14 L1–60	Final taper
W15	L296–309	— (no pack)	—	(W15 note at SE-W14 L58–60)	FINAL + viva

Topic map → where it lives

Topic	Covered in
Process models (Waterfall, Incremental, Spiral, Reuse-oriented)	W1
Requirements engineering + IEEE 830 SRS	W2
DFD / ER / data dictionary / balancing	W3
Project planning, WBS, Gantt, CPM/PERT, RMMM	W4
Design concepts, architectural styles, cohesion/coupling	W5
GoF design patterns, Agile, Scrum, XP	W6
Verification & validation, inspection, testing fundamentals	W9
White-box testing, cyclomatic complexity	W10
Black-box testing (EP/BVA), SQA, metrics	W11
COCOMO (basic/intermediate/detailed), EAF	W12
Maintenance types, reverse vs re-engineering	W13

> **Gaps are flagged, not fabricated:** no `02-Official-Syllabus.md` (topic list derived from narrative headings), no `Lab-Schedule.md`, no `03-Lab-Breakdowns/`. The topic→week map above is derived from `Week-by-Week-Narrative.md` + `weeks/README.md`.

Exam probability table (editorial — study prioritization only)

Topic	Midterm	Final	Numerical	Diagram	Theory	Definition
Process models	90%	50%	0%	20%	50%	30%
Requirements	80%	40%	0%	20%	50%	30%
DFD / Analysis	75%	45%	0%	50%	30%	20%

Project mgmt (WBS, CPM)	65%	40%	30%	30%	25%	15%
Design concepts	50%	65%	0%	30%	45%	25%
Design patterns	40%	70%	0%	30%	45%	25%
Agile/Scrum	50%	60%	0%	10%	55%	35%
White-box testing	0%	75%	30%	25%	30%	15%
Black-box testing	0%	65%	20%	20%	40%	20%
COCOMO	0%	80%	50%	10%	25%	15%
Maintenance	0%	50%	0%	10%	55%	35%

GPA priority: COCOMO ▥ Must Win (numerical, predictable, ~15% of final) · Design patterns ■ · Testing white-box (cyclomatic = easy marks) ■ · Process models ■ · DFD/UML ■ · Agile/Scrum ▨ · Maintenance ▧ (source: `01-Course-Overview.md`).

Marks allocation

Component	Weight	Strategy
COCOMO numericals	~15%	Memorize coefficients, drill 10 problems
DFD/UML diagrams	~15%	Practice with modeling tools
Process models	~15%	Comparison table memorization
Testing (cyclomatic)	~10%	Formula + control flow graphs
Theory	~25%	Structure: definition + explanation + example
Lab project	~40% official (~20% study-effort)	Group management protocol, start early

> **Weight conflict:** `01-Course-Overview.md` L17 lists lab as ~20% *study-effort*; the official split is **60% theory exam + 40% lab (project + reports + viva)** per `Week-by-Week-Narrative.md` L11. Both kept, labeled; the 40% lab is the official risk component (canonical pick: **40% official**).

Exam stacks

- **Midterm (W8, 05–11 Oct):** past paper 60 min → blank page 30 min → error log 20 min. Answer DFD/process-model theory first, theory second. Sleep 8 h. Ledger frozen.
- **Final (W15, 23–29 Nov):** COCOMO numericals → DFD/cyclomatic → theory. For theory questions use the 5-point structure (definition, explanation, example, advantage, disadvantage). Sleep 9 h (banked from W12–W14).
- **Viva (W15, separate slot):** `SE-LABS.md` viva + `Viva-Book.md` — 3-min project walkthrough (problem → design → implementation → testing → lessons learned), then 10 technical questions. Know your SRS and UML diagrams cold.

Resources

YouTube playlists

Playlist	Channel	Videos	Why
[Software Engineering (Complete)]	Gate Smashers	62 (confirmed 2026-08-02)	All topics: process models, SRS, design, testing, SQA, project management
[Software Engineering Full Course]	SECourses	15 lessons / 16.5h	Agile, Scrum, UML breadth
[Java Design Pattern Complete Tutorial]	codeonedigest	count unverified	GoF patterns with Java examples

Textbooks

- **Pressman & Maxim** — *Software Engineering: A Practitioner's Approach*, 9th Ed (2019; McGraw-Hill). ISBN **978-1-260-54800-6**. Process models, requirements, analysis, design, testing, SQA, PM, COCOMO.

- **Ian Sommerville** — *Software Engineering*, 10th Ed (2016; Pearson). ISBN **978-0-13-394303-0**. Agile methods, dependability, maintenance.

Free MOOCs

Course	Source	Notes
NPTEL: Software Engineering	IIT Kharagpur	Prof. Rajib Mall, 12 weeks
Coursera: Software Development Lifecycle	Univ. of Minnesota	SDLC specialisation
edX: Software Engineering: Introduction	UBC	Intro course

Problem banks / practice

Resource	Link
GeeksforGeeks SE Interview Questions	https://www.geeksforgeeks.org/software-engineering/software-engineering-interview-questions-and-answers/
Sanfoundry SE Questions	https://www.sanfoundry.com/software-engineering-questions-answers/
Refactoring Guru: Design Patterns Catalog	https://refactoring.guru/design-patterns/catalog

Cheat sheets / revision notes

Resource	Link
Refactoring Guru: Design Patterns Cheat Sheet	https://refactoring.guru/design-patterns
RIT SWEN-383 Design Patterns RefCard (PDF)	https://www.se.rit.edu/~swen-383/resources/RefCardz/designpatterns.pdf
GitHub: OneMoreGres/patterns-cheatsheet (PDF)	https://github.com/OneMoreGres/patterns-cheatsheet

Secret weapon

Refactoring.Guru — https://refactoring.guru — the most beautifully explained design patterns resource: real-world analogies, UML diagrams, pros/cons, code in 9 languages. Makes GoF patterns actually click, free to use online.

Where-everything-lives index (source → embedded location)

Source file	~Lines	Embedded in
`01-Course-Overview.md`	46	NAV facts/weights + exam probability + marks allocation
`02-Official-Syllabus.md`	—	**absent for SE** (gap, flagged)
`03-Lab-Breakdowns/…`	—	**absent for SE** (gap, flagged)
`Definition-Book.md`	32	per-week "Definitions" blocks (verbatim, 14 entries)
`Formula-Book.md`	25	per-week "Formulas" blocks (verbatim)
`Numerical-Book.md`	14	per-week "Numericals" numbered types (#16–21)
`Diagram-Book.md`	25	per-week "Diagrams" numbered refs (#1–20)
`Marks-Allocation.md`	15	NAV (this file)
`Resources.md`	44	NAV (this file) + per-week book/video lines
`Fear-Killer-Packs.md`	70	per-week packs (verbatim, with per-question offsets)
`Week-by-Week-Narrative.md`	319	per-week narrative (verbatim)
`Lab-Schedule.md`	—	**absent for SE** (gap, flagged)
`Lab-Resources.md`	129	SE-LABS per-lab Resources lines + derived map
`Viva-Book.md`	25	SE-LABS viva section (verbatim)
`weeks/SE-W1…W14` + `README`	801	per-week anchors, P0 floor, drill targets

`Top-10-Tricky-Concepts.md` + `Top-100-Questions.md` are **deferred** for SE (decision 2026-08-04) — equivalent coverage lives in `Definition-Book.md`, `Fear-Killer-Packs.md`, and the topical books. `SE_CS302_lab_.pdf` is unreadable → lab content sourced from `Lab-Resources.md`.

Generated from the SE pack (CS-326). Every offset verified against `weeks/README.md`, `Week-by-Week-Narrative.md`, `Fear-Killer-Packs.md`, and the source files themselves.

Schedule at a glance

W	Dates	Variant	File	Topics	Lab
1	17–23 Aug	P2	BEFORE	Software process models	Lab 1 (proposal)
2	24–30 Aug	P2	BEFORE	Requirements engineering + IEEE 830	Lab 1 (SRS)
3	31 Aug–06 Sep	P2	BEFORE	Analysis modelling — DFD & ER	Lab 4 (DFD)
4	07–13 Sep	P0	BEFORE	Project management — WBS, Gantt, CPM, RMMM	Lab 2 (ProjectLibre)
5	14–20 Sep	P0	BEFORE	Software design & architecture	Labs 3+6 (arch)
6	21–27 Sep	P2	BEFORE	Design patterns & Agile/Scrum	Lab 7 (pattern)
7	28 Sep–04 Oct	P1	BEFORE	Midterm revision	— (no milestone)
8	05–11 Oct	MIDTERM	BEFORE	Exam week — no new material	—
9	12–18 Oct	P1	AFTER	Verification & validation (recovery)	Lab 8 (test design)
10	19–25 Oct	P2	AFTER	White-box testing — cyclomatic	— (no milestone)
11	26 Oct–01 Nov	P2	AFTER	Black-box testing & SQA	Lab 8 (execute tests)
12	02–08 Nov	P2	AFTER	COCOMO — cost estimation	— (COCOMO report)
13	09–15 Nov	P1	AFTER	Software maintenance	— (report finalization)
14	16–22 Nov	P0	AFTER	Final taper — past paper FIRST	— (final submission)
15	23–29 Nov	FINAL	AFTER	Exam + lab viva — execute	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week. Variants are editorial tier assignments mirroring AI/CA/PS — SE manifests carry no variant field. Lab column is **derived** from `weeks/README.md` project milestones (no `Lab-Schedule.md`); workbook Labs 9–13 have no narrative milestone mapped. W8/W15 no lab.